

input + output mapping

Field of study which give computers ability to learn with out being explicitly programmed

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Supervised → data with output

→ Predicts continuous numeric values

Regression: - estimates relationship b/w variables

- used when dependent var is continuous

output is continuous out is also continuous

Li¹ predefined

→ deals with assigning class to input data

→ Classification: - values are categorical or discrete

- output is discrete - assign input data to predefined class / category

e.g. # email is spam or non spam & patient has disease or no disease

- Buy insurance

Classifi Cat² algo²sr

Logistic Regr (for classification prob)

- similar / Extension of Li. regression

Same as Li reg but Logistic or Sigmoid activation

fun¹ which convert it into 0 or 1 (binary)

Binary class

Two class

Multi class

more than 2

(both reg & class) builds (combination problem)

Decision Tree is a flow-chart like model to make predictions

- constructs tree like structure (predicts output based on

- Hierarchical structure (greedy search approach)

- Internal node represent feature / attribute, branch represent possible outcomes and leaves (end node)

represents the predicted / target values

Darsi
Notes

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Random Forest

- Combines d Trees to create more accurate

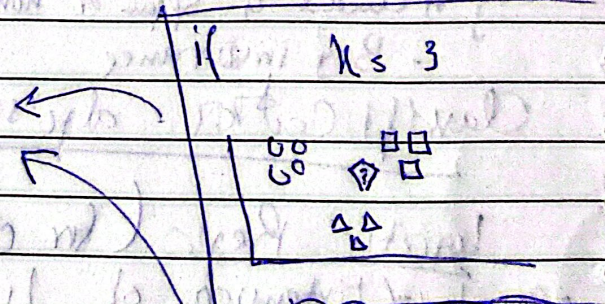
Naive Bayes's (Classification)

- based on probabilistic Bayes theorem
- e.g spam, sentiment analysis
- Naive means that every feature is independent of each other

Bagging
where each dTree is trained on random sample of train data

KNN

- First what is value of k



K means

- Start with $k=2$ which will place two Centroids at random & which is near to it is labeled the class
- Compute distance from Centroid of every point & cluster them accordingly
- Repeat step 2
- We use elbow for correct value of k
- Calc each MSE & SE for each point class and then add two classes & draw plot for the no's
- The elbow will be the no of clusters

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'Confusion Matrix' used to evaluate performance of Classifier model

	P	N
P	TP	FP
N	FN	TN
	Actual	

True Positive
 ↓ actual ↓ predicted

Classification Report used for supervised ML

- it holds values for Precision, recall, F1-Score

1

P_n

For Regression Evaluating Models:

MARE is Error = Actual V - Pred Value

① Precision measure of positive predictions (FP)

$$\frac{TP}{TP+FP}$$

② Recall out of all positive pred How much are actual, How much correctly identified

F1 $\rightarrow$ mean of ① & ②

$$\frac{TP}{TP+FN}$$

Darshi
Notes

Deep Learning: Specialized subset of ML which focuses on the development & training of artificial NN to perform complex tasks & learn from data.

Activation functions: They determine output of a neuron in NN

- Step Function: output 1 if > 1 or 0 otherwise
 - Sigmoid: if less than 0.5 out is 0, if greater outputs 1
(for binary classification)
 - Tanh: Output b/w -1 & +1
(for hidden layers)
 - ReLU: if input is -ve it out 0, +ve output without modification
(popular choice in hidden layers)
 - Leaky ReLU: same for +ve no but for -ve output close to zero
 - Softmax: transform input into proba distribution which equals to 1
(used in last layer of multiclass classification)
- or cost function

Loss function: It guides optimization algo to adjust network parameters (weights & biases) to minimize loss.

- MSE: Regression problem
- Binary Cross Entropy: Binary problems (two classes)
- Categorical & S: multi class classification problem
- Sparse & S & S: same as categorical (C) but it handles integers instead of One Hot encoded one.

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→ adjust weights & biases

Ques:- Optimizer:- primary goal → minimize the loss function
improve performance by updating network parameters.
- Adam
- SGD

Metrics:- Evaluation metrics during testing & training
- Accuracy - F1, recall, Precision - mAP

CNN:-